

Lecture 7:

Application Layer

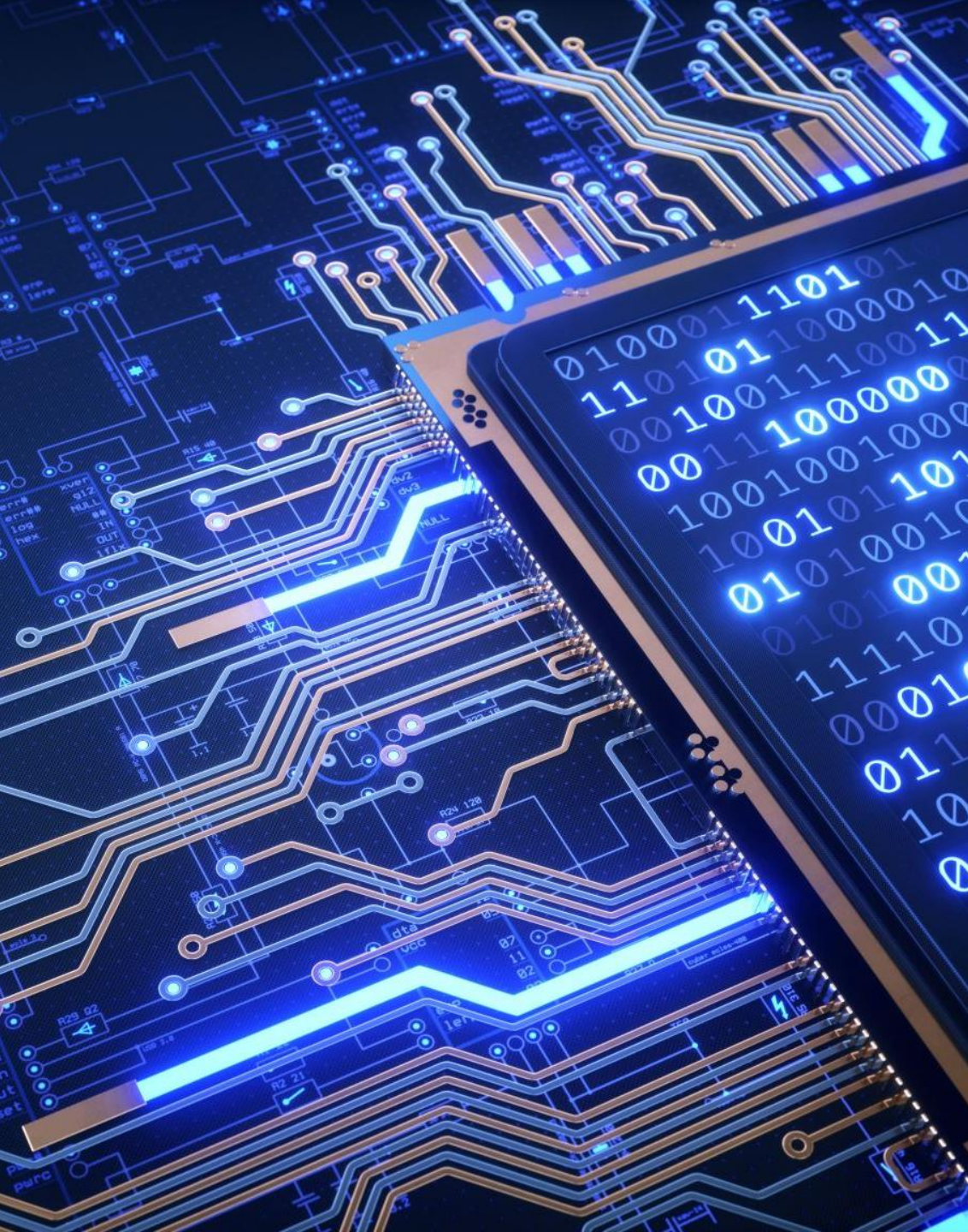
Protocol Ep.2 :

DNS

R wiangsripanawan

05.01.26

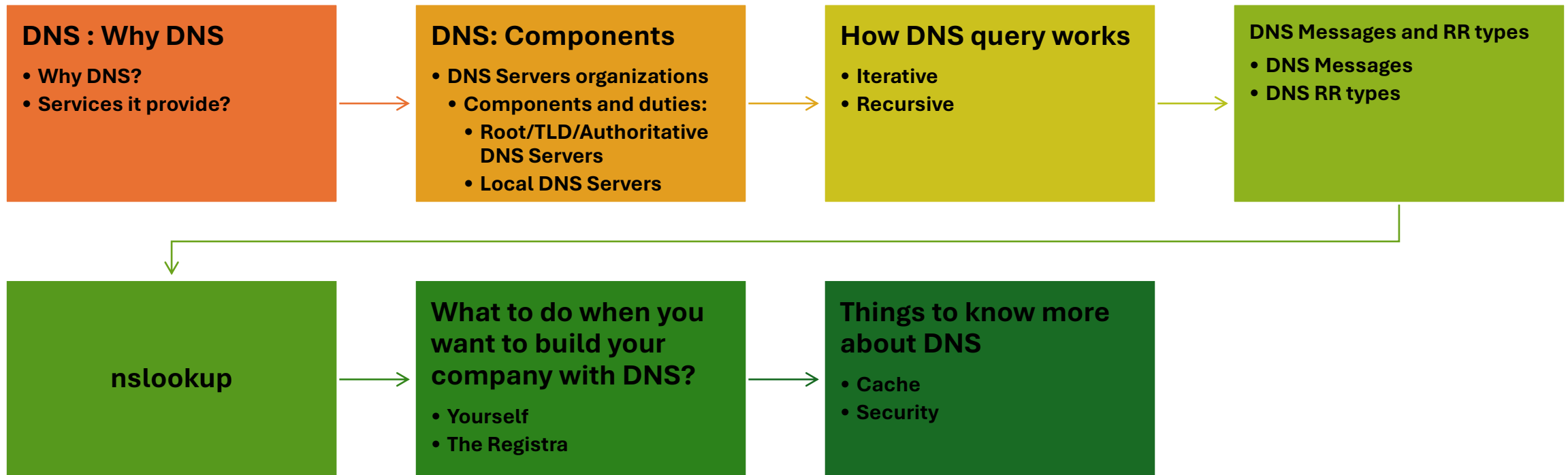




Outcome

- Understand the importance of DNS
- Understand DNS Name servers; type and roles (Root DNS, TLD DNS, Authoritative DNS and local DNS)
- Understand DNS query mechanisms both iterate and recursive
- Understand DNS Message: DNS Query and DNS Reply
- Understand DNS RR Types
- Being able and understand how to use command nslookup
 - to query IP address (given hostname) and understand its return values.
 - To query hostname (given IP address)
 - To query Authoritative DNS Servers
 - To query mail servers
- Being able to understand what to set in your organization RR record when you start building your Authoritative DNS Server.

Outline



DNS



Slide page 63 - 79



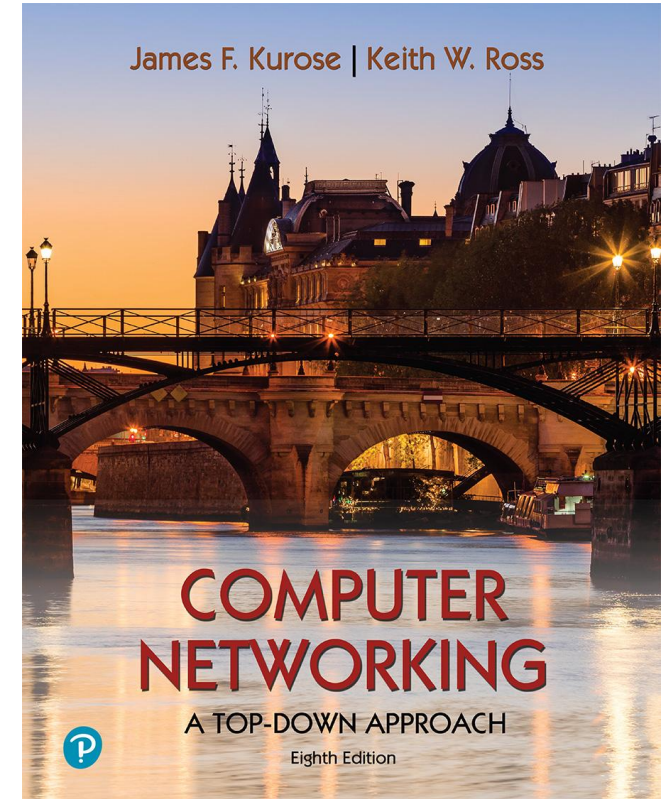
นำมาจาก **Powerpoint slides**
Chapter 2: Application Layer

*Page 2-1 – 2-53 Principles of
network applications*



**Book : Computer Networking: A
Top-Down Approach 8th edition**
โดย **Jim Kurose, Keith Ross,**
Pearson, 2020

https://gaia.cs.umass.edu/kurose_ross/ppt.php



*Computer Networking: A
Top-Down Approach*

8th edition

Jim Kurose, Keith Ross
Pearson, 2020

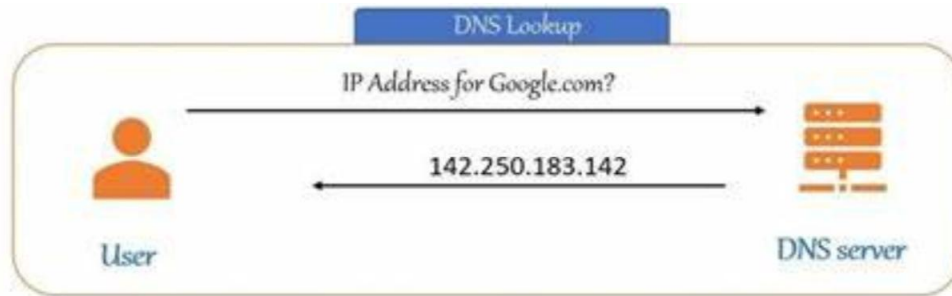
Why DNS?



- Computers communicate via IP Address.
- **BUT !!!** คนจำหมายเลข IP ของเครื่อง เช่น server ที่จะติดต่อด้วยไม่ได้
- จริงๆ แล้ว OS จะมีไฟล์เก็บ mapping ของ ชื่อ Host กับ IP เช่น
 - Windows : c:\windows\system32\drivers\etc\hosts.file
 - Unix : /etc/resolve
 - แต่การ update host file บ่อยๆ ไม่ work โดยเฉพาะการใช้งานใน Internet
- เลยต้องมีตัวช่วย => **Domain Name System**
 - แทนที่ host จะเก็บ mapping เอง แต่ละ host จะไปถามเครื่องที่ให้บริการการค้นหาชื่อให้
 - เรียกเครื่องที่ให้บริการนี้ว่า Domain Name System Server (DNS Server)

1. Which one do you think it should be like?

A



DNS server เครื่องหนึ่งเก็บ ชื่อ Host หมดทั่วโลก

B



As clients or users we cannot see it but there actually is the system workings on name service

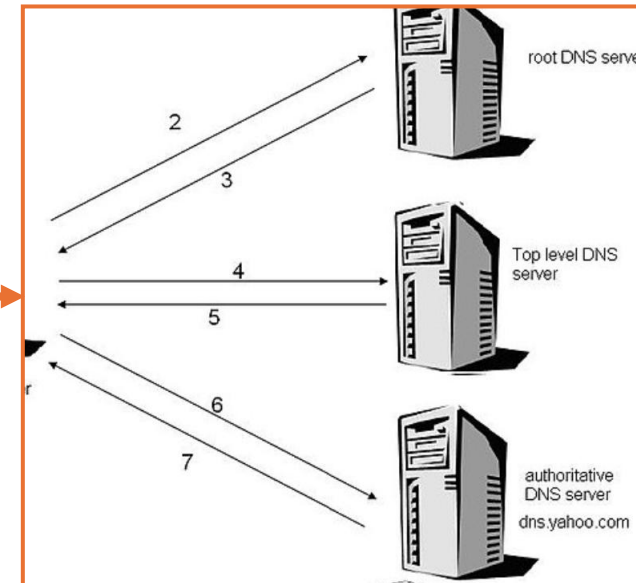
2. Which one do you think the DNS infrastructure should be like?



A Centralize Database one big global DNS Server



B Distributed Database



Why not centralize DNS?

Q: Why not centralize DNS?

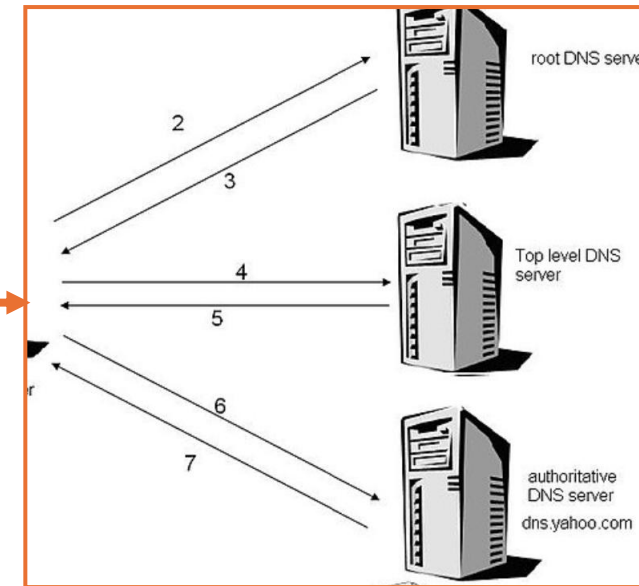
- single point of failure
- traffic volume
- distant centralized database
- maintenance

A: doesn't scale!

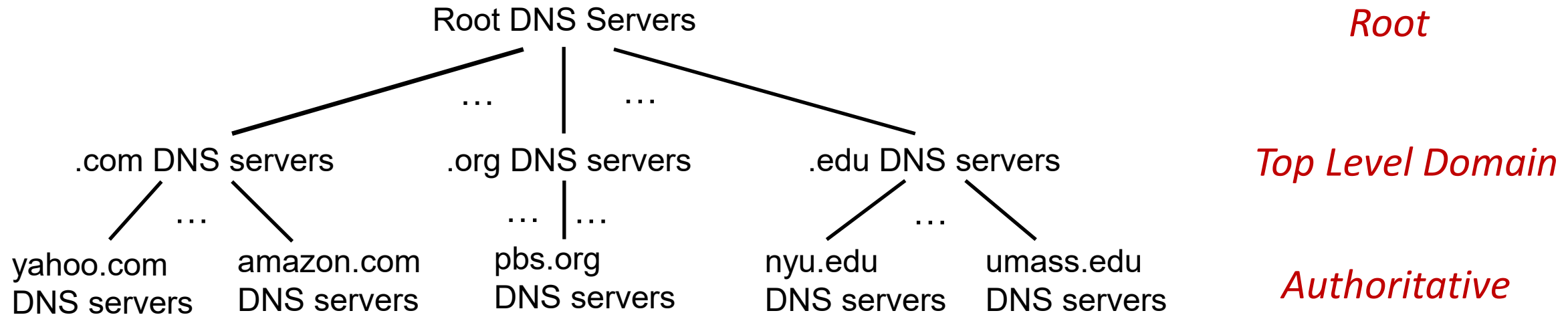
- Comcast DNS servers alone: 600B DNS queries/day
- Akamai DNS servers alone: 2.2T DNS queries/day



B Distributed Database



DNS: a distributed, hierarchical database

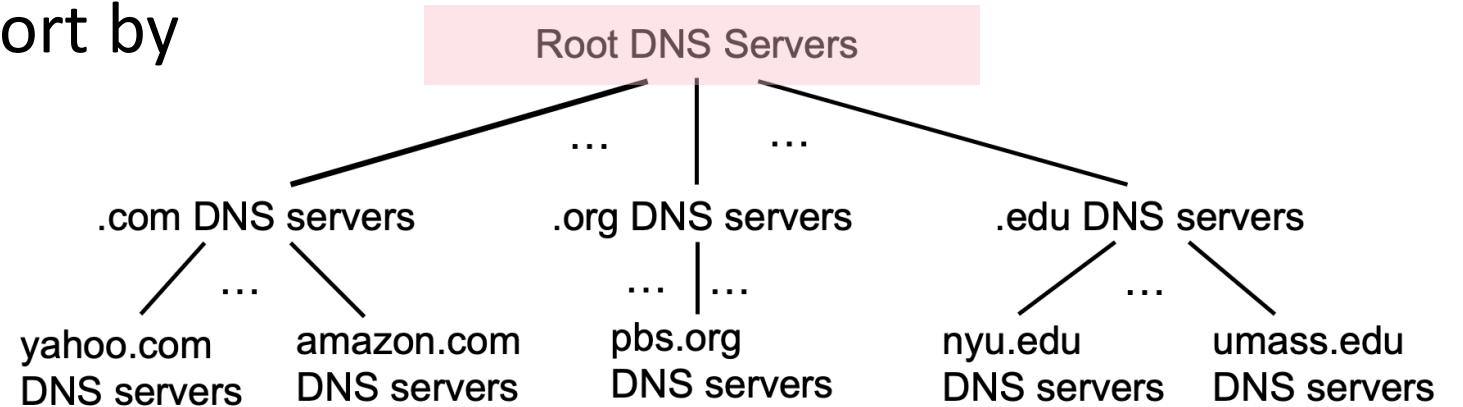


Client wants IP address for www.amazon.com; 1st approximation:

- client queries root server to find .com DNS server
- client queries .com DNS server to get amazon.com DNS server
- client queries amazon.com DNS server to get IP address for www.amazon.com

DNS: root name servers

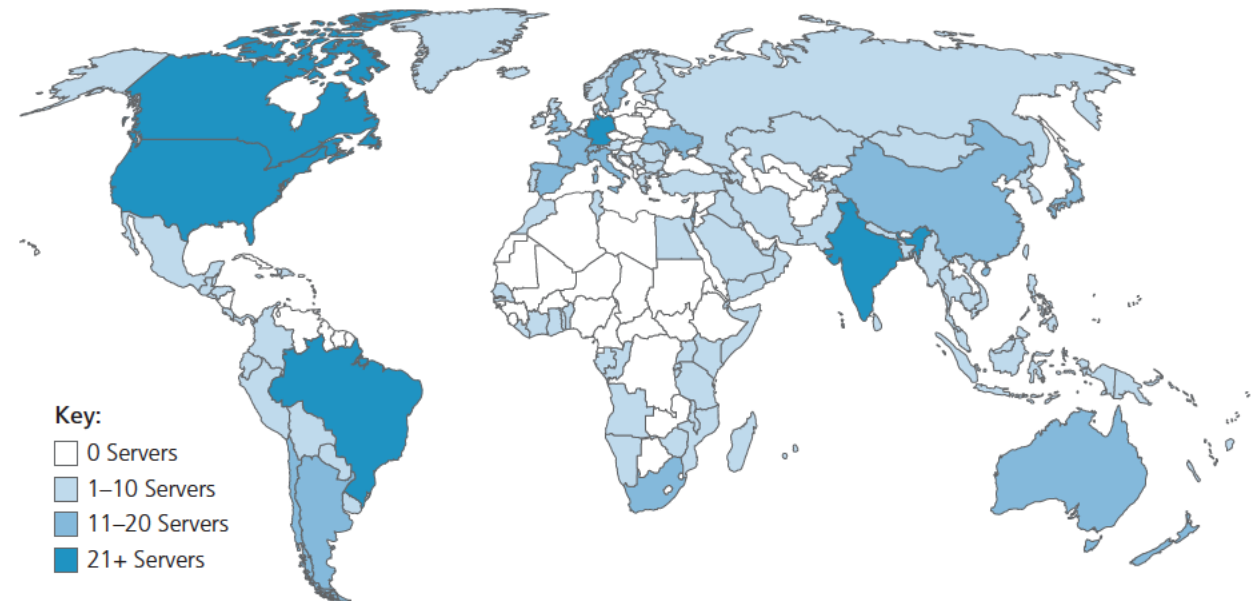
- official, contact-of-last-resort by name servers that can not resolve name



DNS: root name servers

- official, contact-of-last-resort by name servers that cannot resolve name
- *incredibly important* Internet function
 - Internet couldn't function without it!
 - DNSSEC – provides security (authentication, message integrity)
- ICANN (Internet Corporation for Assigned Names and Numbers) manages root DNS domain

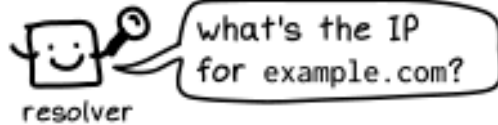
13 logical root name “servers”
worldwide each “server” replicated
many times (~200 servers in US)



JULIA EVANS
@bork

the root nameservers

every DNS resolver
starts at the
root nameservers



You should ask a .com
nameserver!
It's at a.gtld-servers.net.



root nameserver

root nameserver IP
addresses almost
never change

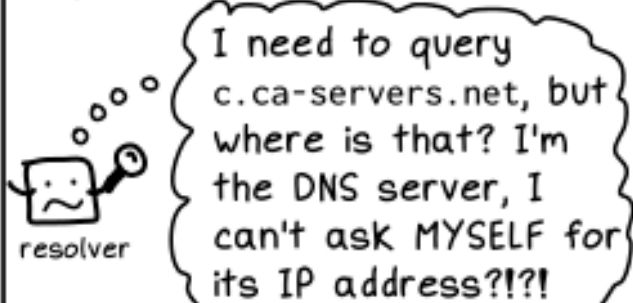
a.root-servers.net's IP
(198.41.0.4) hasn't changed
since 1993 ← **DECADES**
ago!

there are thousands of
physical servers, but
only 13 IP addresses

Each IP refers to multiple
physical servers, you'll get
the one closest to you

There's a map at root-servers.org

if they didn't exist,
resolvers wouldn't know
where to start



every resolver has the
root IPs hardcoded in
its source code

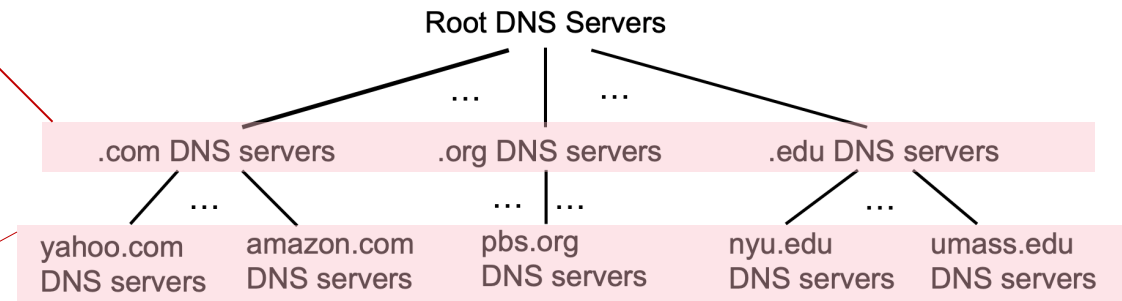
Here they are!
You can try making DNS
queries to them :)

a.root-servers.net	198.41.0.4
b.root-servers.net	199.9.14.201
c.root-servers.net	192.33.4.12
d.root-servers.net	199.7.91.13
e.root-servers.net	192.203.230.10
f.root-servers.net	192.5.5.241
g.root-servers.net	192.112.36.4
h.root-servers.net	198.97.190.53
i.root-servers.net	192.36.148.17
j.root-servers.net	192.58.128.30
k.root-servers.net	193.0.14.129
l.root-servers.net	199.7.83.42
m.root-servers.net	202.12.27.33

Top-Level Domain, and authoritative servers

Top-Level Domain (TLD) servers:

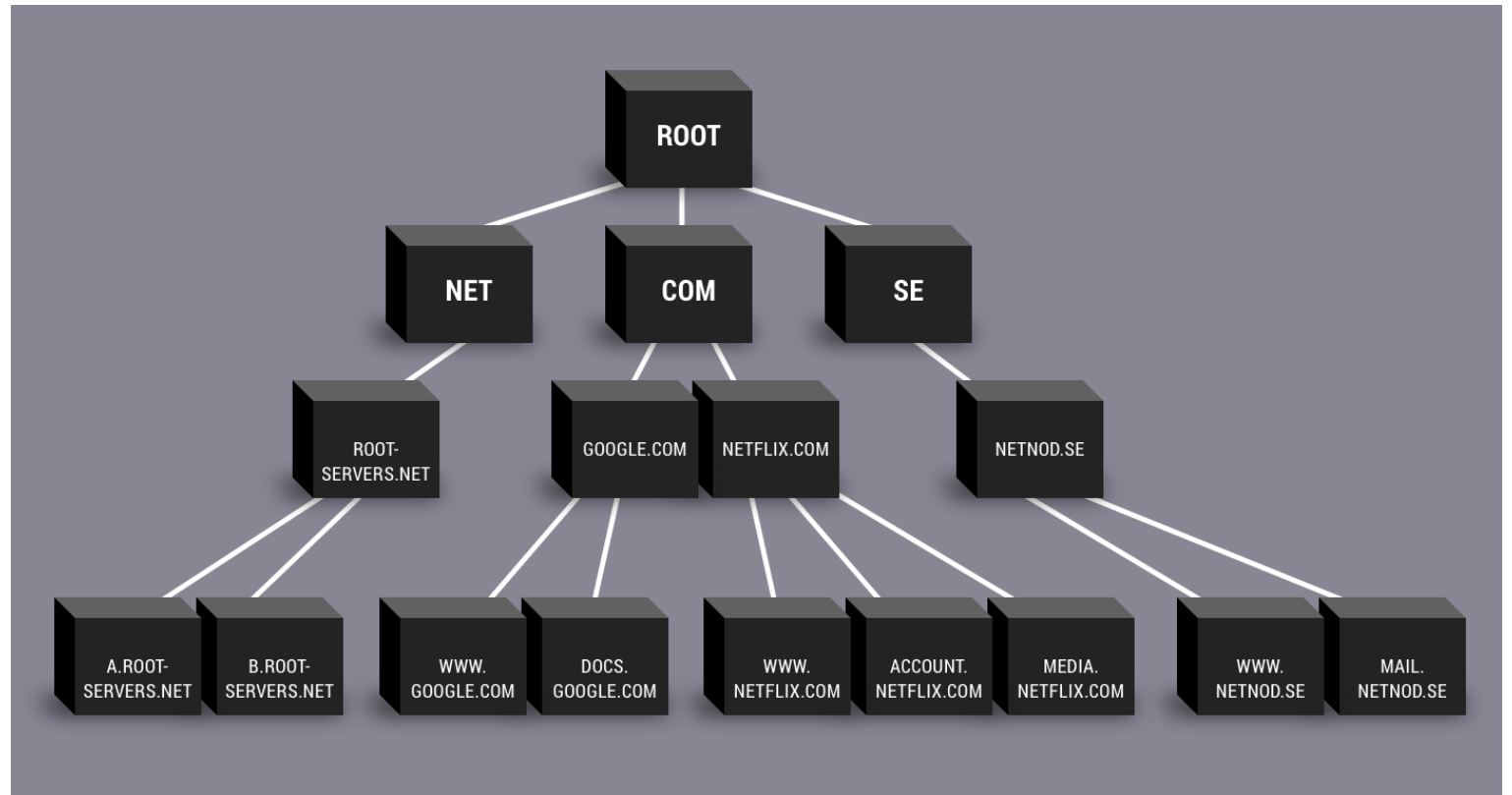
- responsible for .com, .org, .net, .edu, .aero, .jobs, .museums, and all top-level country domains, e.g.: .cn, .uk, .fr, .ca, .jp
- Network Solutions: authoritative registry for .com, .net TLD
- Educause: .edu TLD



authoritative DNS servers:

- organization's own DNS server(s), providing authoritative hostname to IP mappings for organization's named hosts
- can be maintained by organization or service provider

Recap: เครื่องใดเป็น
root dns server,
TLD dns servers,
authoritative dns
servers and
hosts.



DNS: Domain Name System

Domain Name System (DNS):

- *distributed database*
implemented in hierarchy of many *name servers*
- *application-layer protocol*: hosts, DNS servers communicate to *resolve* names (address/name translation) (client – server model)
 - *note*: core Internet function, implemented as application-layer protocol
 - complexity at network's "edge"

DNS services:

- hostname-to-IP-address translation
- host aliasing
 - canonical, alias names
- mail server aliasing
- load distribution
 - replicated Web servers: many IP addresses correspond to one name

Local DNS name servers

- when host makes DNS query, it is sent to its *local* DNS server
 - Local DNS server returns reply, answering:
 - from its local cache of recent name-to-address translation pairs (possibly out of date!)
 - forwarding request into DNS hierarchy for resolution
 - each ISP has local DNS name server; to find yours:
 - MacOS: `% scutil --dns`
 - Windows: `>ipconfig /all`
- local DNS server doesn't strictly belong to hierarchy

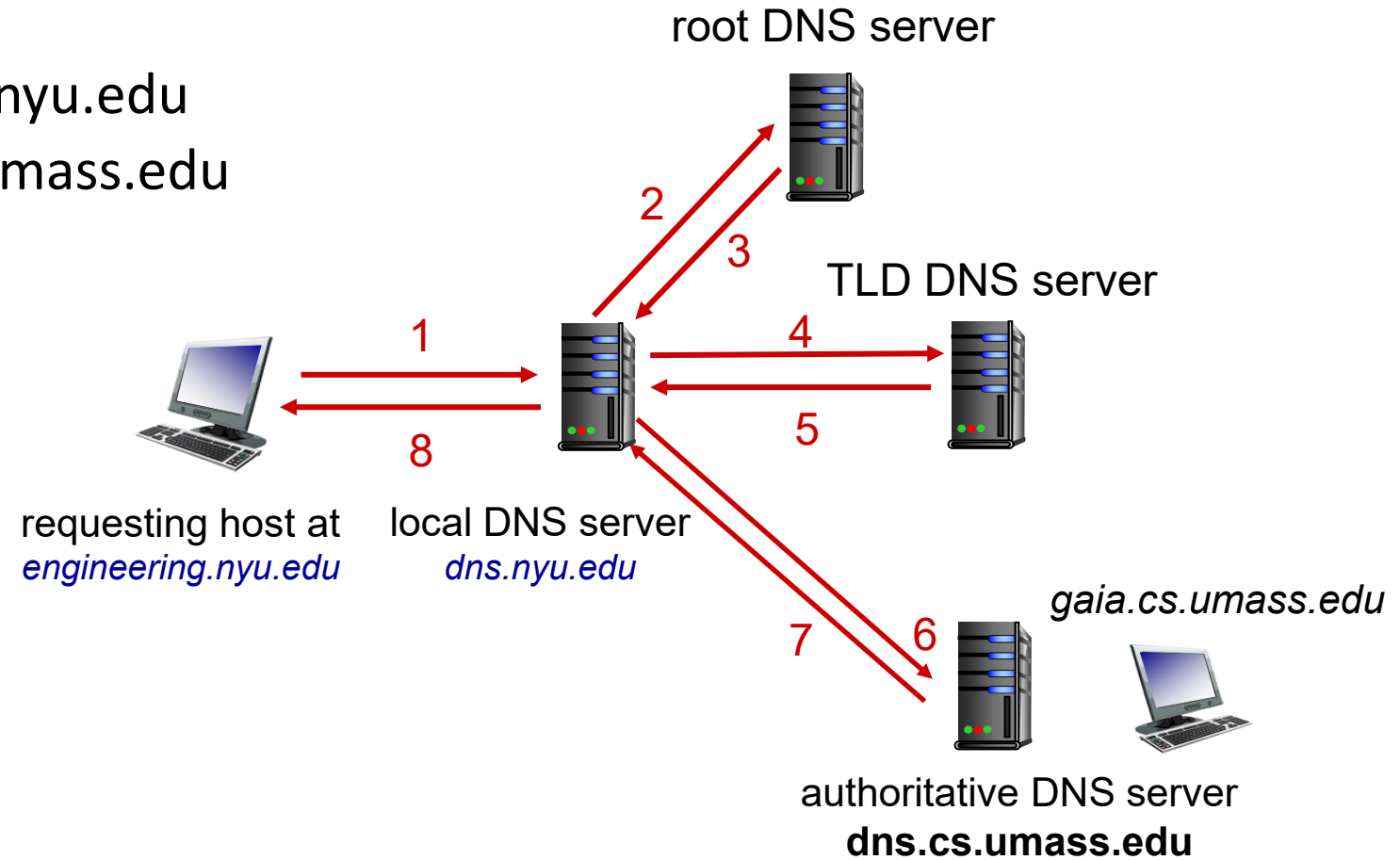


DNS name resolution: iterated query

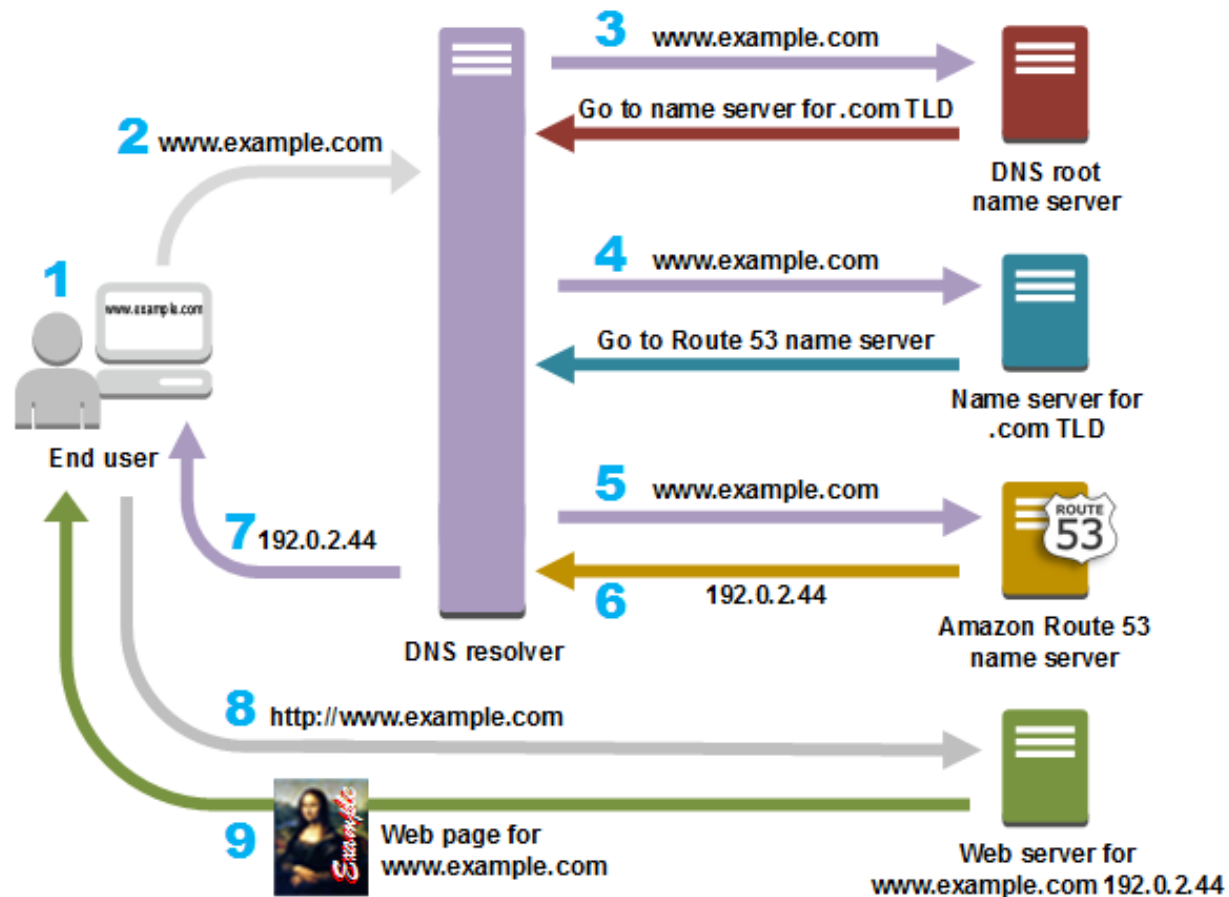
Example: host at `engineering.nyu.edu` wants IP address for `gaia.cs.umass.edu`

Iterated query:

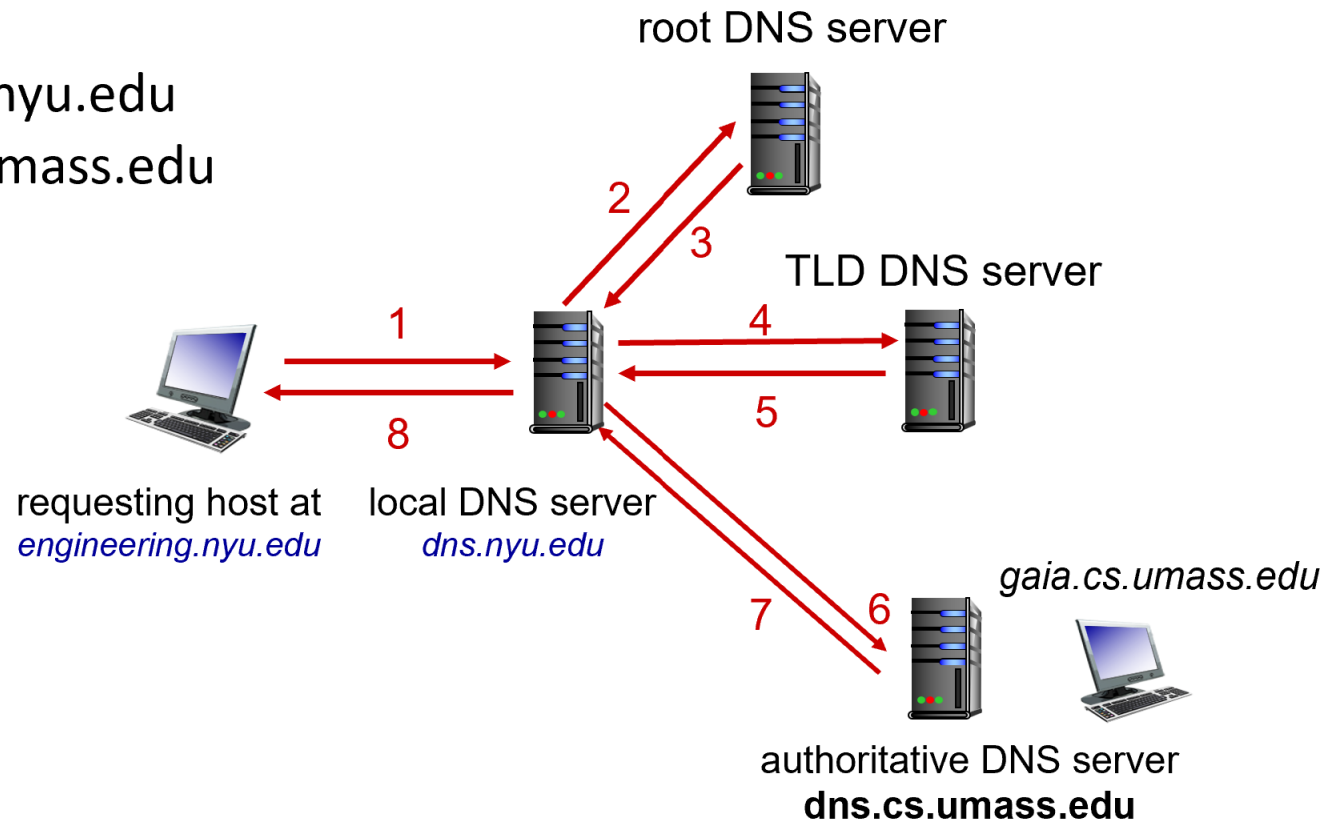
- contacted server replies with name of server to contact
- “I don’t know this name, but ask this server”



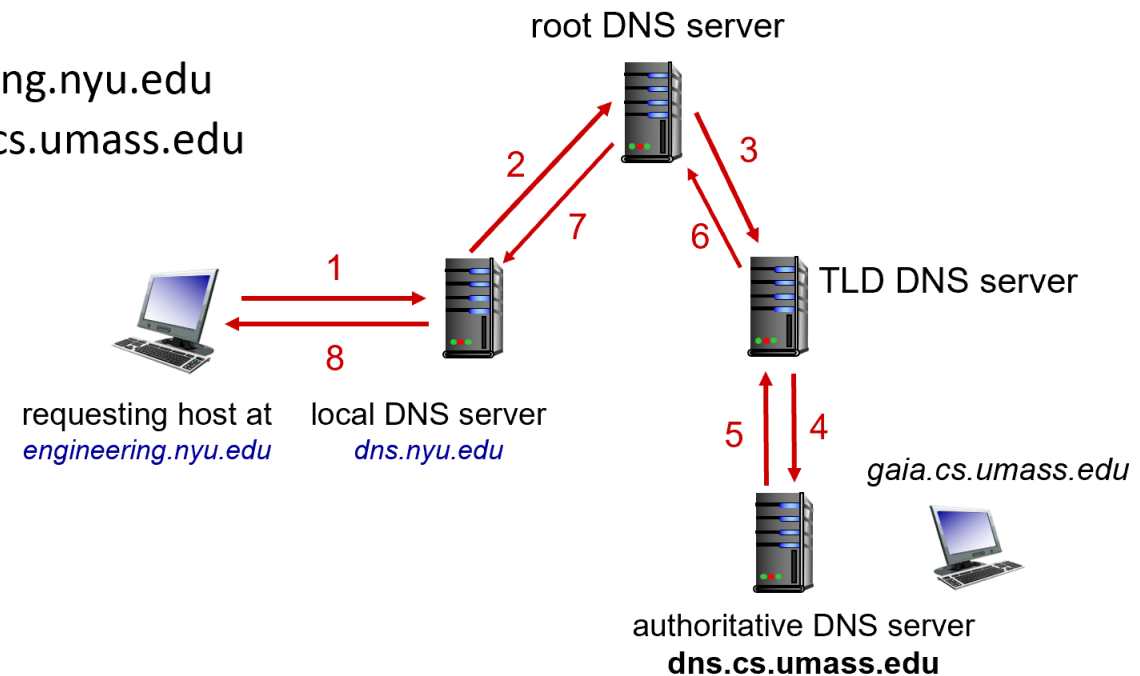
ภาพนี้มาจากเว็บไซต์ AWS ถ้ายังไม่เข้าใจลองทำความเข้าใจดูนะคะเป็น Iterated



nyu.edu
mass.edu



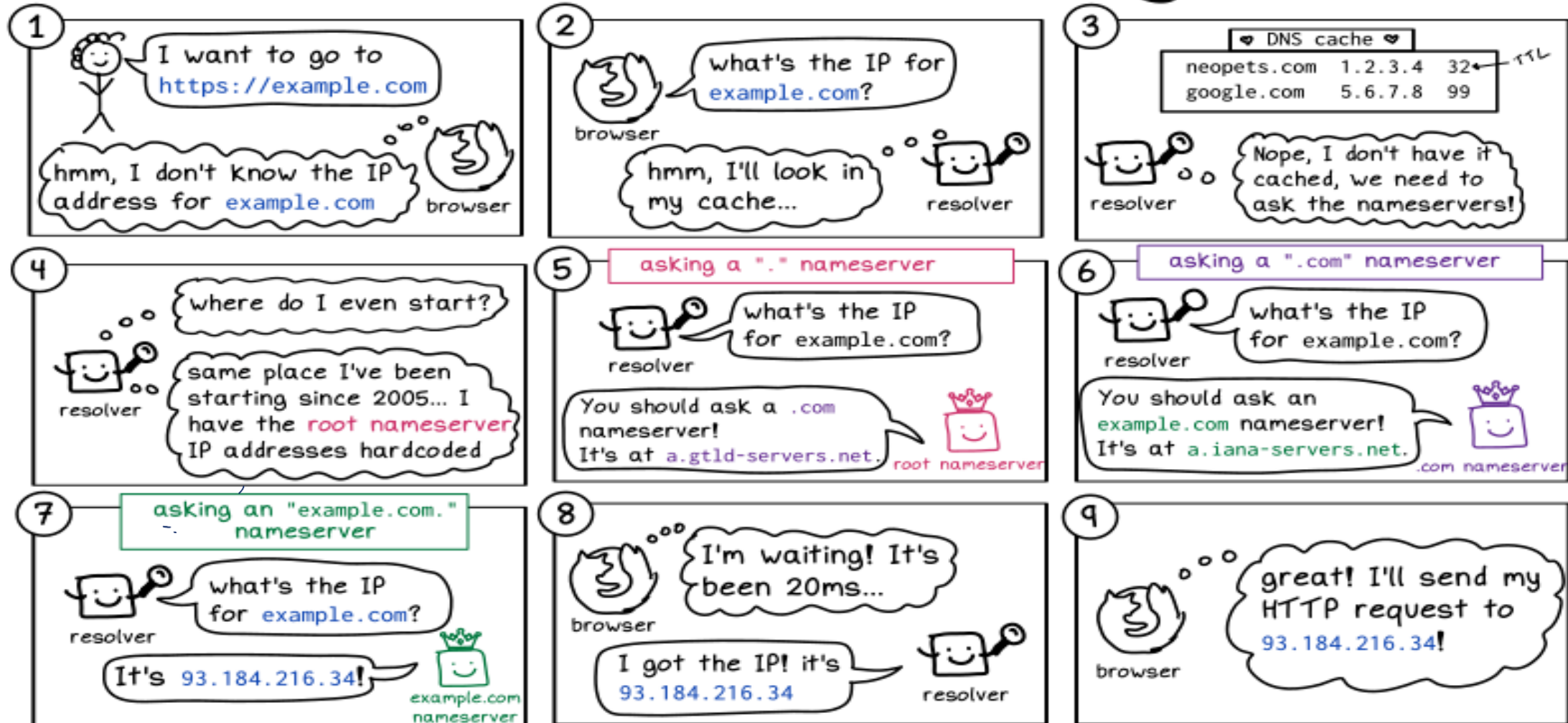
ng.nyu.edu
cs.umass.edu



ถ้ายังไม่เข้าใจอีกลองอ่านการ์ตูนนี้ดูค่ะ

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@bork

life of a DNS query



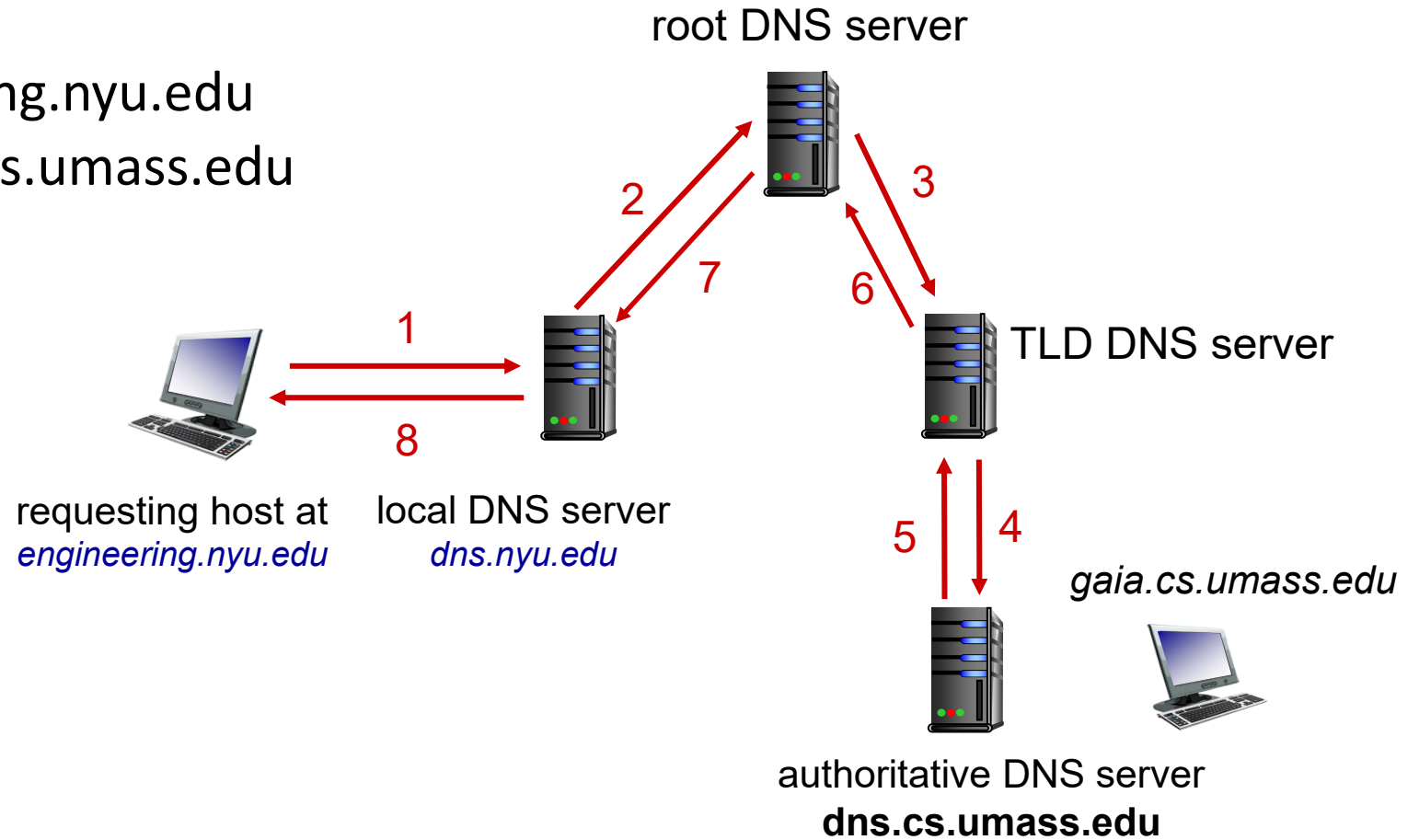
<https://pbs.twimg.com/media/FJ6GmPuXMAIx4Xw.png>

DNS name resolution: recursive query

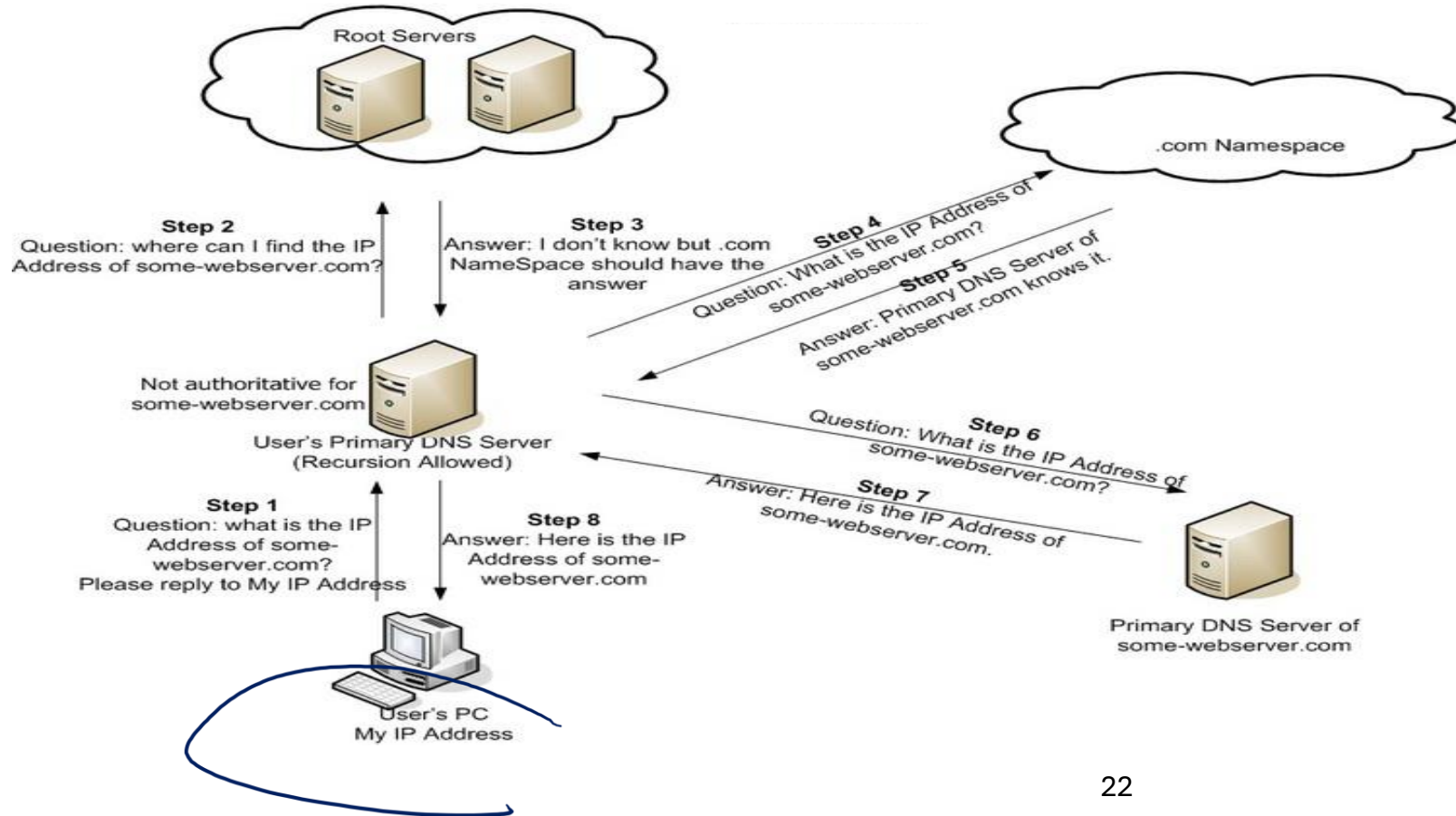
Example: host at `engineering.nyu.edu` wants IP address for `gaia.cs.umass.edu`

Recursive query:

- puts burden of name resolution on contacted name server
- heavy load at upper levels of hierarchy?



นักศึกษาคิดว่าจากรูปเป็นการ query แบบใด



Question : นักศึกษาคิดว่าส่วนใหญ่จะ implement DNS name resolver แบบไหน
เพราะอะไร



ITERATIVE OR RECURSIVE



WHY?



Let's have a look at what
are inside the database?

DNS records

DNS: distributed database storing resource records (RR)

RR format: (name, value, type, ttl)

type=A

- name is hostname
- value is IP address

type=NS

- name is domain (e.g., foo.com)
- value is hostname of authoritative name server for this domain

type=CNAME

- name is alias name for some “canonical” (the real) name
- www.ibm.com is really servereast.backup2.ibm.com
- value is canonical name

type=MX

- value is name of SMTP mail server associated with name

Caching DNS Information

- once (any) name server learns mapping, it *caches* mapping, and *immediately* returns a cached mapping in response to a query
 - caching improves response time
 - cache entries timeout (disappear) after some time (TTL)
 - TLD servers typically cached in local name servers
- cached entries may be *out-of-date*
 - if named host changes IP address, may not be known Internet-wide until all TTLs expire!
 - *best-effort name-to-address translation!*

How to use nslookup

```
C:\Users\acer>nslookup www.facebook.com
Server:  UnKnown
Address:  172.20.10.1

Non-authoritative answer:
Name:     star-mini.c10r.facebook.com
Addresses: 2a03:2880:f18b:188:face:b00c:0:25de
           163.70.149.35
Aliases:  www.facebook.com
```

```
C:\Users\acer>nslookup www.facebook.com 1.1.1.1
Server:  one.one.one.one
Address:  1.1.1.1

Non-authoritative answer:
Name:     star-mini.c10r.facebook.com
Addresses: 2a03:2880:f18b:188:face:b00c:0:25de
           163.70.149.35
Aliases:  www.facebook.com
```

```
C:\Users\acer>nslookup www.facebook.com 8.8.8.8
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:     star-mini.c10r.facebook.com
Addresses: 2a03:2880:f18b:82:face:b00c:0:25de
           163.70.148.35
Aliases:  www.facebook.com
```

Nslookup แปล ข้อมูล **RR Record**
จาก **DNS Reply** ออกมาในรูปแบบที่ผู้ใช้
คำสั่งสามารถเข้าใจได้ แต่ไม่ได้แสดง **TTL**

```
C:\Users\acer>nslookup www.thairath.co.th
Server: UnKnown
Address: 172.20.10.1
```

```
Non-authoritative answer:
Name: www.thairath.co.th
Addresses: 3.166.34.87
           3.166.34.43
           3.166.34.7
           3.166.34.116
```

```
C:\Users\acer>nslookup pantip.com
Server: UnKnown
Address: 172.20.10.1
```

```
Non-authoritative answer:
Name: pantip.com
Addresses: 2001:c00:4618:cafe:1996::167
           2001:c00:4618:cafe:1996::168
           2001:c00:4618:cafe:1996::165
           2001:c00:4618:cafe:1996::169
           2001:c00:4618:cafe:1996::166
           203.151.13.166
           203.151.13.165
           203.151.13.168
           203.151.13.169
           203.151.13.167
```

```
C:\Users\acer>nslookup onedrive.com 8.8.8.8
Server: dns.google
Address: 8.8.8.8
```

```
Non-authoritative answer:
Name: onedrive.com
Address: 20.101.246.164
```

```
C:\Users\acer>nslookup onedrive.com 1.1.1.1
Server: one.one.one.one
Address: 1.1.1.1
```

```
Non-authoritative answer:
Name: onedrive.com
Address: 51.105.104.217
```

nslookup Type A

```
C:\Users\acer>nslookup www.science.kmitl.ac.th
Server: UnKnown
Address: 172.20.10.1
```

```
Non-authoritative answer:
Name: www.science.kmitl.ac.th
Address: 161.246.127.150
```

```
C:\Users\acer>nslookup www.kmitl.ac.th
Server: UnKnown
Address: 172.20.10.1
```

```
Non-authoritative answer:
Name: www.kmitl.ac.th
Address: 161.246.127.182
```

```
C:\Users\acer>nslookup e11285.dsca.akamaiedge.net
Server: UnKnown
Address: 172.20.10.1
```

```
Non-authoritative answer:
Name: e11285.dsca.akamaiedge.net
Addresses: 2001:fb0:102d:18f::2c15
           2001:fb0:102d:1a7::2c15
           23.192.101.163
```

type=A

- name is hostname
- value is IP address

```
C:\Users\acer>nslookup google.com
Server: UnKnown
Address: 172.20.10.1
```

```
Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4001:800::200e
           216.58.221.206
```

nslookup กรณี RR เป็นข้อมูลแบบ CNAME

```
C:\Users\acer>nslookup www.huawei.com
Server: UnKnown
Address: 172.20.10.1

Non-authoritative answer:
Name: e11285.dsca.akamaiedge.net
Addresses: 2001:fb0:102d:18f::2c15
           2001:fb0:102d:1a7::2c15
           23.192.101.163
Aliases: www.huawei.com
          www.huawei.com.akadns.net
          ion-sslsv6.huawei.com.edgekey.net
```

```
C:\Users\acer>nslookup www.huawei.com.akadns.net
Server: UnKnown
Address: 172.20.10.1

Non-authoritative answer:
Name: e11285.dsca.akamaiedge.net
Addresses: 2001:fb0:102d:18f::2c15
           2001:fb0:102d:1a7::2c15
           23.192.101.163
Aliases: www.huawei.com.akadns.net
          ion-sslsv6.huawei.com.edgekey.net
```

```
C:\Users\acer>nslookup ion-sslsv6.huawei.com.edgekey.net
Server: UnKnown
Address: 172.20.10.1

Non-authoritative answer:
Name: e11285.dsca.akamaiedge.net
Addresses: 2001:fb0:102d:18f::2c15
           2001:fb0:102d:1a7::2c15
           23.192.101.163
Aliases: ion-sslsv6.huawei.com.edgekey.net
```

type=CNAME

- name is alias name for some “canonical” (the real) name
- www.ibm.com is really servereast.backup2.ibm.com
- value is canonical name

nslookup

Type NS

```
C:\Users\acer>nslookup -type=NS google.com 1.1.1.1
Server:   one.one.one.one
Address:  1.1.1.1
```

Non-authoritative answer:

```
google.com      nameserver = ns2.google.com
google.com      nameserver = ns1.google.com
google.com      nameserver = ns3.google.com
google.com      nameserver = ns4.google.com
```

```
C:\Users\acer>nslookup -type=NS google.com 8.8.8.8
Server:  dns.google
Address: 8.8.8.8
```

Non-authoritative answer:

```
google.com      nameserver = ns1.google.com
google.com      nameserver = ns4.google.com
google.com      nameserver = ns2.google.com
google.com      nameserver = ns3.google.com
```

```
C:\Users\acer>nslookup -type=NS kmitl.ac.th
Server:  UnKnown
Address: 172.20.10.1
```

Non-authoritative answer:

```
kmitl.ac.th     nameserver = diamond.ce.kmitl.ac.th
kmitl.ac.th     nameserver = clarinet.asianet.co.th
kmitl.ac.th     nameserver = ns1.kmitl.ac.th
```

```
C:\Users\acer>nslookup -type=NS true.co.th
Server:  UnKnown
Address: 172.20.10.1
```

Non-authoritative answer:

```
true.co.th      nameserver = mario.ns.cloudflare.com
true.co.th      nameserver = isla.ns.cloudflare.com
```

type=NS

- name is domain (e.g., foo.com)
- value is hostname of authoritative name server for this domain

nslookup Type MX

type=MX

- value is name of SMTP mail server associated with name

```
C:\Users\acer>nslookup -type=MX kmitl.ac.th
Server: UnKnown
Address: 172.20.10.1

Non-authoritative answer:
kmitl.ac.th      MX preference = 10, mail exchanger = alt3.aspmx.l.google.com
kmitl.ac.th      MX preference = 5, mail exchanger = alt2.aspmx.l.google.com
kmitl.ac.th      MX preference = 10, mail exchanger = alt4.aspmx.l.google.com
kmitl.ac.th      MX preference = 11, mail exchanger = mailsecure1.kmitl.ac.th
kmitl.ac.th      MX preference = 5, mail exchanger = alt1.aspmx.l.google.com
kmitl.ac.th      MX preference = 1, mail exchanger = aspmx.l.google.com
```

```
C:\Users\acer>nslookup -type=MX true.co.th
Server: UnKnown
Address: 172.20.10.1

Non-authoritative answer:
true.co.th      MX preference = 10, mail exchanger = mx02.mail.icloud.com
true.co.th      MX preference = 10, mail exchanger = mx01.mail.icloud.com
```

```
C:\Users\acer>nslookup -type=MX chula.ac.th
Server: UnKnown
Address: 172.20.10.1

Non-authoritative answer:
chula.ac.th     MX preference = 20, mail exchanger = mail2-gw1.it.chula.ac.th
chula.ac.th     MX preference = 100, mail exchanger = mail-gw1.it.chula.ac.th
```

```
C:\Users\acer>nslookup -type=MX cmu.ac.th
Server: UnKnown
Address: 172.20.10.1

Non-authoritative answer:
cmu.ac.th       MX preference = 0, mail exchanger = cmu-ac-th.mail.protection.outlook.com
```

```
C:\Users\acer>nslookup -type=MX psu.ac.th
Server: UnKnown
Address: 172.20.10.1

Non-authoritative answer:
psu.ac.th       MX preference = 0, mail exchanger = psu-ac-th.mail.protection.outlook.com
```

What about DNS message



DNS Query

UDP : Client Port (not fixed)



DNS Reply

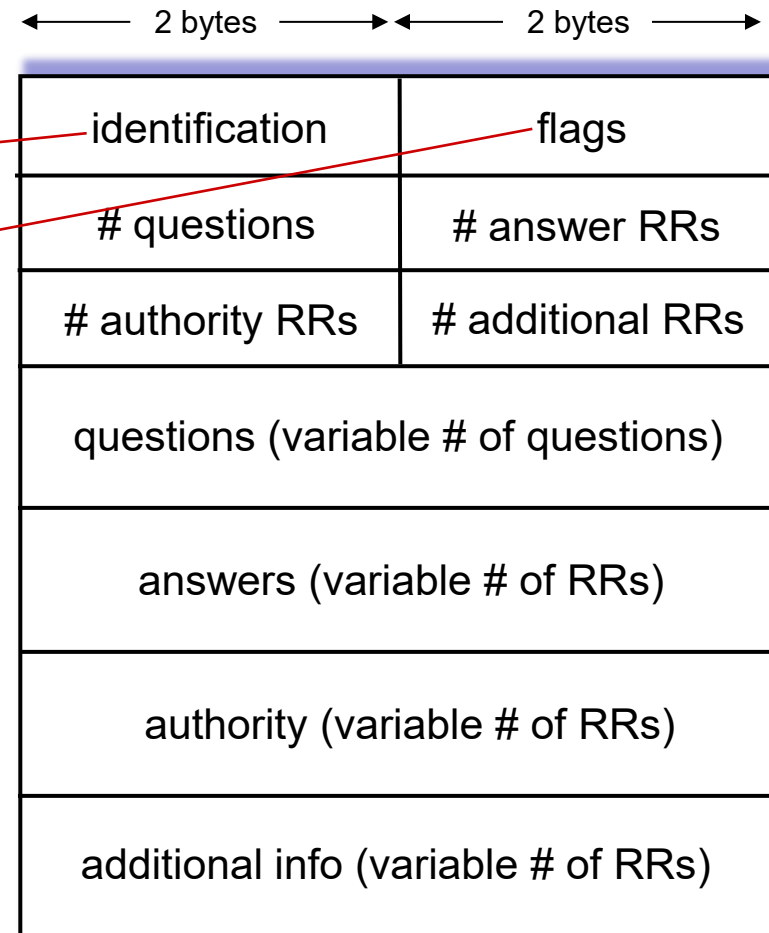
UDP:Server Port 53

DNS protocol messages

DNS *query* and *reply* messages, both have same *format*:

message header:

- **identification**: 16 bit # for query, reply to query uses same #
- **flags**:
 - query or reply
 - recursion desired
 - recursion available
 - reply is authoritative



DNS protocol messages

DNS *query* and *reply* messages, both have same *format*:

← 2 bytes → ← 2 bytes →

identification	flags
# questions	# answer RRs
# authority RRs	# additional RRs
questions (variable # of questions)	
answers (variable # of RRs)	
authority (variable # of RRs)	
additional info (variable # of RRs)	

name, type fields for a query

RRs in response to query

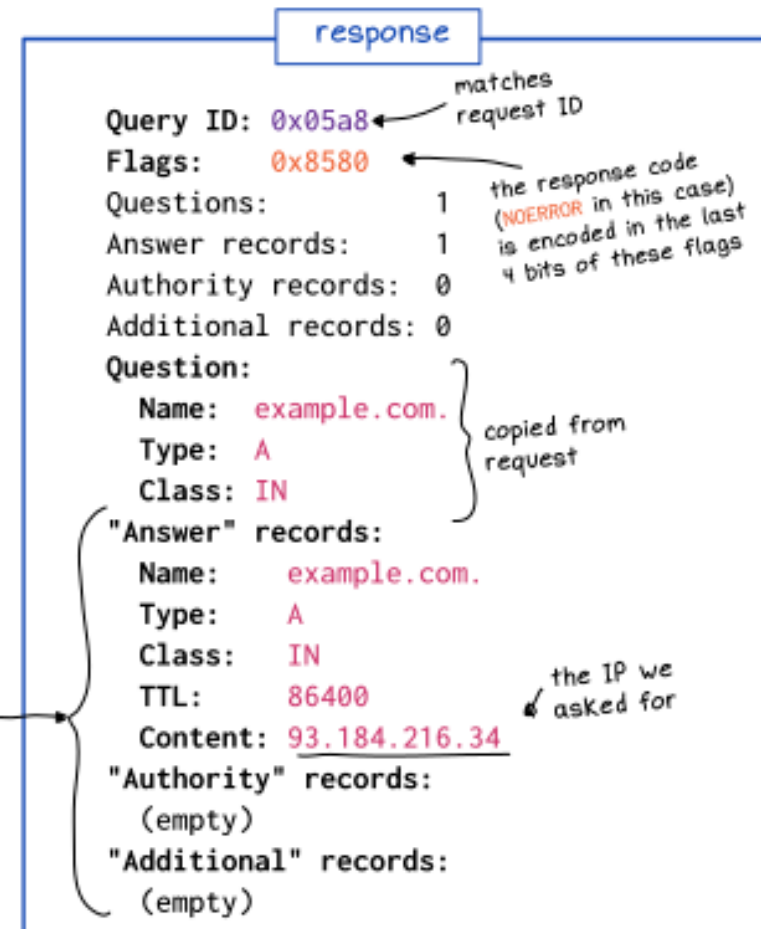
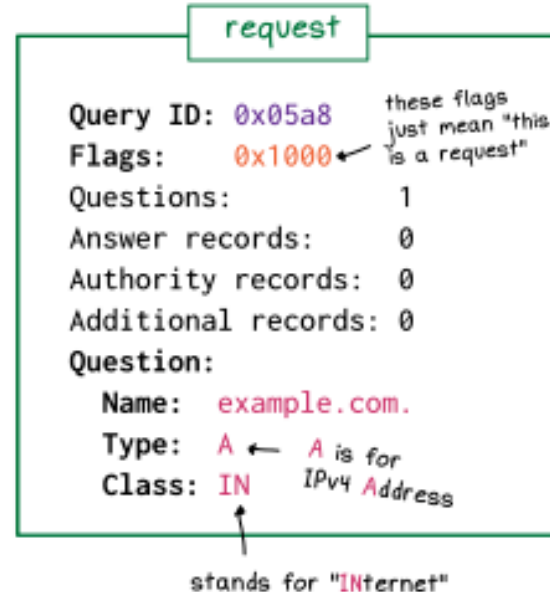
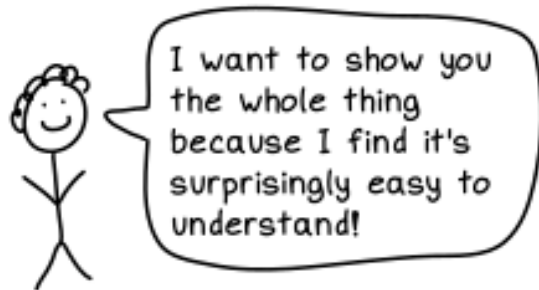
records for authoritative servers

additional “helpful” info that may
be used

everything inside a DNS packet

I literally mean everything, I copied this verbatim from a real DNS request in Wireshark

Let's look at the actual data being sent during this interaction!



Getting your info into the DNS

example: new startup "Network Utopia"

1. Us -> Registrar: register name **networkutopia.com** at *DNS registrar* (e.g., Network Solutions)

- provide names, IP addresses of *authoritative name server (primary and secondary)*

2. Registrar -> with its TLD server : **inserts NS, A RRs into .com TLD server**: # (Registrar ทำการเพิ่มโดเมนเนมของบริษัทของเราลงไปใน TLD server database ตามด้วย Record ของ Authoritative Name server ขององค์กรเรา)

❑(networkutopia.com, dns1.networkutopia.com, NS)

❑(dns1.networkutopia.com, 212.212.212.1, A)

3. Us -> ที่เครื่อง Authoritative Name Server : **create authoritative server locally with IP address 212.212.212.1**

❑Add type A record for www.networkutopia.com

➤ (www.networkutopia.com, 212.212.212.2, A, #ttl)

❑Add type MX record for networkutopia.com ถ้ามี

➤ (networkutopia.com, mail.networkutopia.com, MX, #ttl)

➤ (mail.network.utopia.com, 212.212.212.2, A, #ttl)

Exercise

- นักศึกษาสร้างบริษัท รหัสนักศึกษา.com
- นักศึกษาไปจด IP มาแล้วเรียบร้อยได้ IP เป็น 202.68.156.0/24
- องค์กรของนักศึกษาอย่างน้อยต้องมีเครื่องต่อไปนี้
 - dns.รหัสนักศึกษา.com เป็น dns server
 - www.รหัสนักศึกษา.com เป็น web server
 - dhcp.รหัสนักศึกษา.com เป็น dhcp server
- นักศึกษาจะต้องทำอย่างไรบ้าง เพื่อให้
 - Web server ของ บริษัทนักศึกษา ชื่อ www.รหัสนักศึกษา.com คนใน internet สามารถ query ได้

1. นักศึกษาจะเลือก Registra อย่างไร

ip autho DNS server

2. ต้องแจ้งข้อมูลอะไรบ้างให้ Registra

→ Domain Name

ชื่อ autho DNS server

3. Registra จะต้อง add ข้อมูลอะไรบ้างลงไป
ใน database

123.com, ns1.123.com, NS, 3000

ns1.123.com, 1.2.3.4, A, 3000

4. Authoritative DNS server ของ

นักศึกษา อย่างน้อยต้องมีข้อมูลอะไรบ้าง

123.com, mx.123.com, MX, 3000

mx.123.com, 1.2.3.5, A, 3000

5. ในกรณีต้องการเพิ่ม mail server ที่จัดการ
email ของคนในองค์กร จะต้องทำอย่างไร

Thinking about the DNS

humongous distributed database:

- ~ billion records, each simple

handles many *trillions* of queries/day:

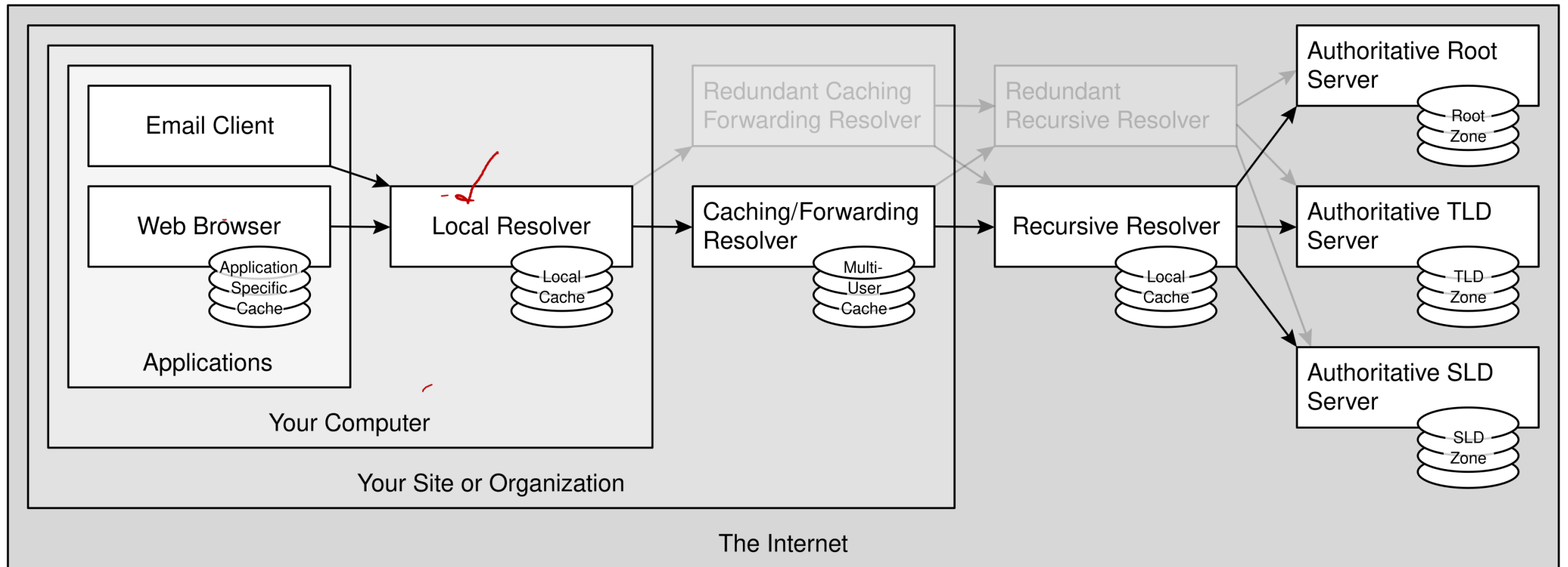
- *many* more reads than writes
- *performance matters*: almost every Internet transaction interacts with DNS - msec count!

organizationally, physically decentralized:

- millions of different organizations responsible for their records

“bulletproof”: reliability, security





DNS security

DDoS attacks

- bombard root servers with traffic
 - not successful to date
 - traffic filtering
 - local DNS servers cache IPs of TLD servers, allowing root server bypass
- bombard TLD servers
 - potentially more dangerous

Spoofing attacks

- intercept DNS queries, returning bogus replies
 - DNS cache poisoning
 - RFC 4033: DNSSEC authentication services

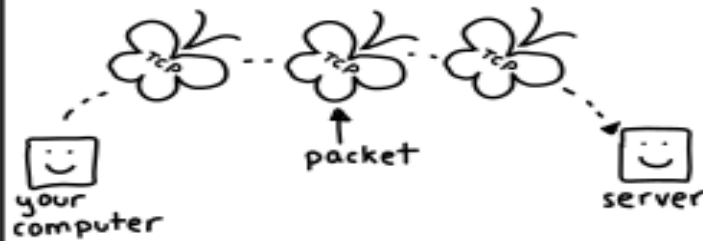
Summary



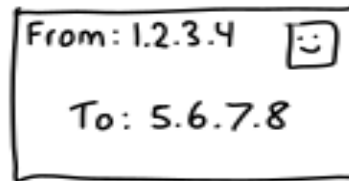
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@bork

why we need DNS

all information on the internet is sent using packets

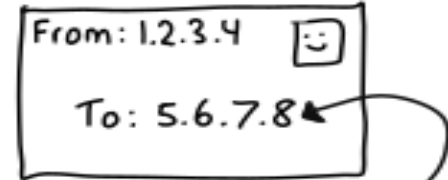


every* network packet has a source and destination IP address



* this is a lie (ARP packets don't) but it's true for 99.9% of packets

network packets aren't sent using domain names



there's nowhere to put a domain name here, just an IP address!

to contact a server, you need to find its IP



DNS was built in 1983 to make it easier to look up IP addresses

before DNS, to update an IP address you needed to make a phone call to a central office

the basics of how DNS works haven't really changed since 1983

it'll probably still work the same way in 30 years too!
so you only need it learn it once :)

how does DNS work?

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more of these at drawings.jvns.ca

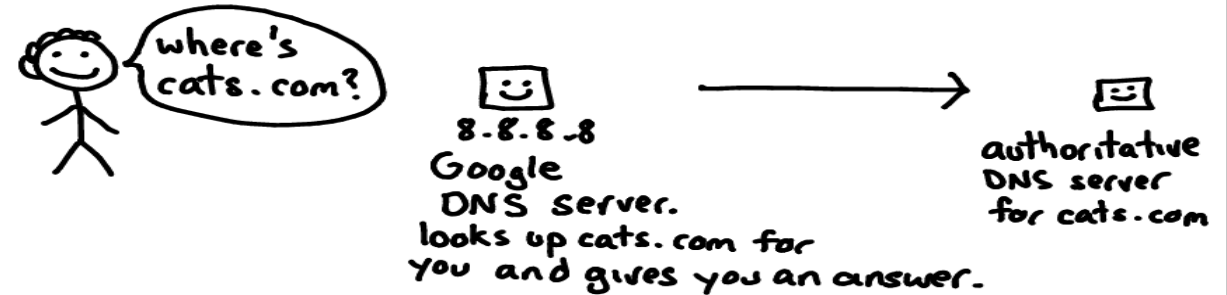
DNS servers translate
names to IP addresses



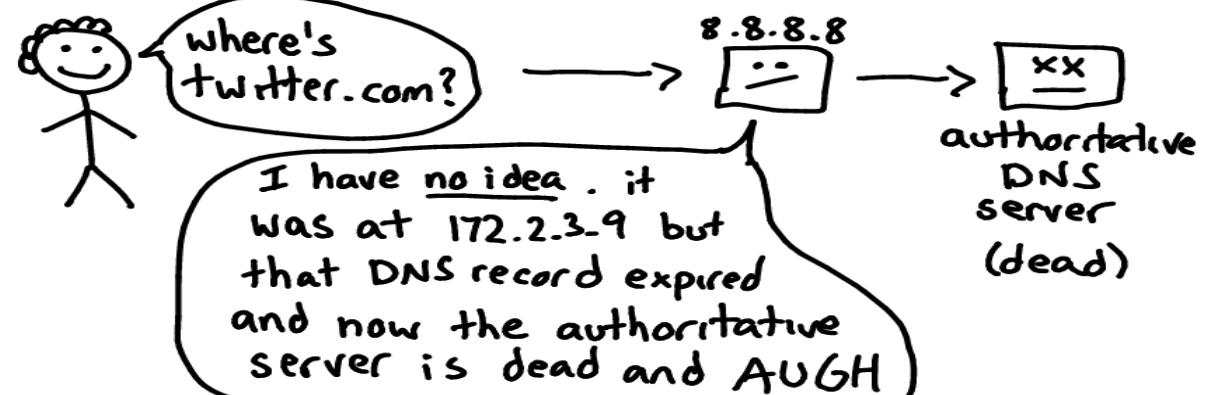
sometimes they tell you it's
an alias (CNAME record)



most DNS queries get cached

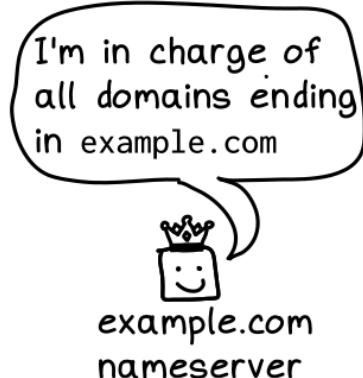
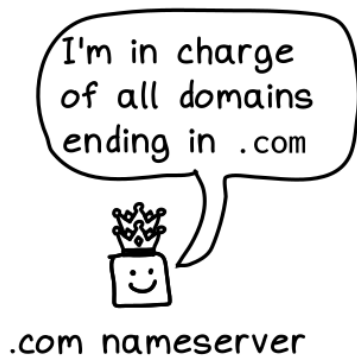
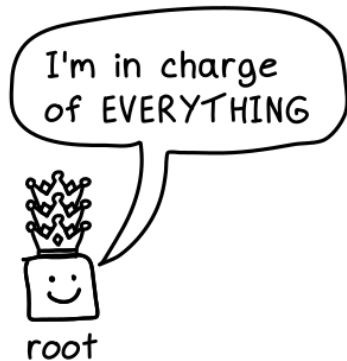


When an important
DNS server dies



the DNS hierarchy

there are 3 main levels of DNS servers



the root nameserver
delegates

what's the IP for example.com?

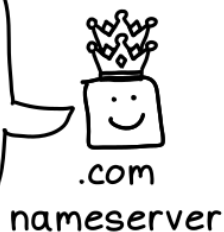
I am not concerned with petty details like that. Here's the address of the *.com nameserver*.



the .com nameserver
also *delegates*

what's the IP for example.com?

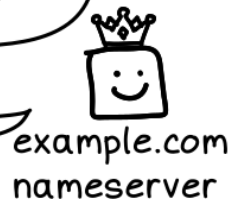
I am not concerned with petty details like that either. Here's the address of the *example.com nameserver*.



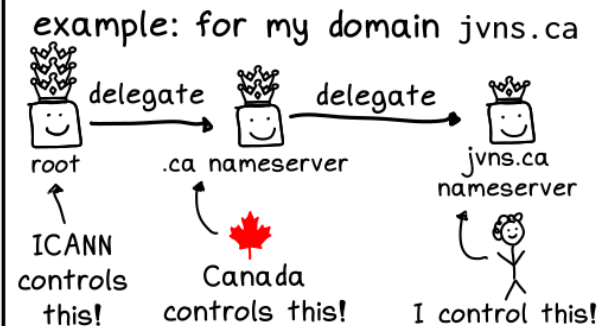
the example.com
nameserver actually
answers your questions

what's the IP for example.com?

93.184.216.34!



this design lets DNS
be decentralized



DNS queries

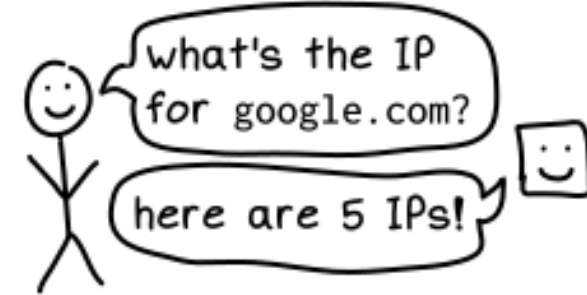
DNS requests are called "queries"



queries have 2 fields

- name (like google.com.)
 - record type (like A)
 - "A" means "IPv4 Address"
 - in DNS, all names end with a dot
- (there's also the "class" but that's literally always set to the same value, "IN" for "internet")

DNS **responses** can contain many **records**



records have 4 or more fields

- name (like google.com.)
- type (like A)
- TTL (like 30)
- one or more content fields (like 172.217.13.110)

responses are organized into 3 **sections**

- "Answer" records
 - Responses to the query (like records with type A) go here
- "Authority" records
 - NS records go here. This section is "I don't know, ask X instead"
- "Additional" records
 - This section contains glue records (which we'll talk about later) and other things.

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why updating DNS takes time

so you updated your domain's IP address

bananas.com	A ▼	300	1.2.3.4
↑ your domain name	↑ record type	↑ TTL in seconds	↑ old IP
bananas.com	A ▼	300	5.6.7.8
		↑ new IP	

the TTL (time to live) is how long the old IP gets cached

30 seconds later...
(you go to bananas.com in your browser)

hey what's the IP for bananas.com?

let's check my cache for bananas.com... found it!!

it's 1.2.3.4!

browser

DNS server

45 seconds later...
(you impatiently refresh bananas.com again to see if it's updated yet)

ok! I just looked that up, so I'll use the IP I cached! (1.2.3.4)

your browser has a DNS cache too!

browser

6 minutes later...
(you refresh the page again)

hey what's the IP for bananas.com?

The TTL (5 minutes) is up, better ask for a new IP...

it's 5.6.7.8!

browser

DNS server

2 hours later...
(someone else visits bananas.com)

hey what's the IP for bananas.com?

it's 1.2.3.4!

another browser

misbehaving DNS server

some ISP DNS servers will ignore your TTL and cache everything for longer

nameserver records usually have higher TTLs than A records

this means that updating your nameservers can be a LOT slower than just updating an IP! Check the TTL with:

```
$ dig ns yourdomain.com
```

life of a DNS query

